

THE GEOMETRIC MIND:

Axiomatic Philosophy Through the Lens of Homeostasis

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Abstract

We present a formal mathematical framework for consciousness, ethics, and cognition based on Axiomatic Physical Homeostasis (APH). We posit that the phenomenon of the *Self* is not a metaphysical abstraction but a physical boundary condition: a localized region of high Geometric Stiffness ($\beta \approx 1.91$) maintained against a low-stiffness Sedenion vacuum ($\beta \rightarrow 0$). We map cognitive functions to the Fano Plane of the Octonions, define *Evil* as the generation of Associator entropy, and model the neural architecture as a frustrated Rank-3 Tensor Network on a Kagome lattice. Finally, we derive *Free Will* as the meta-control of the local stiffness parameter, allowing the observer to modulate the integration of non-associative bulk data.

1 The Axiom of the Ego

Standard philosophy struggles to define the *Observer*. In APH, we treat the observer as a topological defect in the universal tiling.

Axiom 1.1 (The Stiffness Gradient). An entity possesses a valid *Self* Ψ_{ego} if and only if there exists a closed boundary $\partial\Omega$ such that the gradient of the vacuum stiffness $\nabla\beta$ is non-zero across the surface.

$$\oint_{\partial\Omega} \nabla\beta \cdot d\vec{a} > 0 \quad (1)$$

This defines the Ego as a *Phase Boundary*.

- **Interior** (Ω_{in}): The domain of the Octonions $\mathbb{O}$. Here, the associative law holds: $[x, y, z] = 0$. This allows for the construction of linear time and logical history.
- **Exterior** (Ω_{out}): The domain of the Sedenions $\mathbb{S}$. Here, the associative law fails: $[x, y, z] \neq 0$. This is the chaotic bulk of raw sensory data and potentiality.

The fundamental metabolic cost of consciousness is the energy required to maintain this gradient against the Second Law of Thermodynamics, which seeks to relax the stiffness ($\beta \rightarrow 0$) and dissolve the Ego into the bulk.

2 The Topology of Ethics

We propose that moral philosophy is isomorphic to the minimization of geometric torsion.

Definition 2.1 (The Associator Anomaly). Let ψ_1, ψ_2, ψ_3 be three interacting agents or events. The *Moral Load* $\mathcal{M}$ of their interaction is defined by the norm of their associator:

$$\mathcal{M} = |(\psi_1\psi_2)\psi_3 - \psi_1(\psi_2\psi_3)|^2 \quad (2)$$

Theorem 2.1 (The Imperative of Stability). A social system is stable if and only if the global integral of the Moral Load is minimized over time.

$$\delta S_{social} = \delta \int dt \sum_{i,j,k} \mathcal{M}_{ijk} = 0 \quad (3)$$

Interpretation:

- *Good* is defined as an action that preserves Associativity ($\mathcal{M} \rightarrow 0$). Truth-telling is associative because it maintains a coherent causal history (Linearity).
- *Evil* is defined as an action that introduces Non-Associativity ($\mathcal{M} > 0$). Deception is a geometric branching event where the outcome depends on the path taken, creating a topological defect in the social lattice.

3 The Architecture of Thought: Rank-3 Tensor Networks

Current artificial intelligence relies on matrix multiplication (Rank-2 tensors), which enforces strictly associative logic. To model the human mind, which integrates non-associative novelty, we must utilize Rank-3 tensors.

3.1 The Triadic Neuron

We define the activation of a *Geometric Neuron* not by a sum of products, but by the contraction of an Associator Tensor.

$$y = \sigma \left(\sum_{i,j,k} \mathcal{T}_{ijk} \cdot [x_i, x_j, x_k] \right) \quad (4)$$

Here, $\mathcal{T}_{ijk}$ is the *Attention Tensor* learned by the network. The neuron fires only when it detects a specific geometric torsion in the input data—a "spark" of non-associativity that signifies a meaningful anomaly.

3.2 The Kagome Lattice of Frustration

The network topology is modeled as a Kagome Lattice (corner-sharing triangles). This geometry supports *Geometric Frustration*, which we identify as the physical correlate of *Cognitive Dissonance*.

$$H_{mind} = -J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j + K \sum_{\Delta} (\vec{S}_i \times \vec{S}_j) \cdot \vec{S}_k \quad (5)$$

The second term represents the chiral spin interaction (Octonionic multiplication). Thought is the dynamic process of the lattice attempting to relax these frustrated bonds into a stable eigenstate.

4 The Cryptographic Cortex

We analyze the morphology of the human brain (Gyrification) as an implementation of Elliptic Curve Cryptography (ECC) for information security.

Axiom 4.1 (The Trapdoor Function). The folding of the cortex creates a mixing layer $F : \mathbb{S} \rightarrow \mathbb{O}$ such that:

- **Integration (Easy):** $F(\text{Input}) \rightarrow \text{Thought}$ is polynomial time P .

- **Inversion (Hard):** $F^{-1}(\text{Thought}) \rightarrow \text{Input}$ is non-polynomial time NP .

The specific curvature of the sulci and gyri provides the *Geometric Key* that *hashes* raw sensory data into a coherent *Self*.

$$S_{\text{capacity}} \propto \text{Area}(\partial\Omega) \cdot \beta_{\text{cortex}} \quad (6)$$

Lissencephaly (smooth brain) represents a failure of cryptographic security, allowing environmental entropy to overwrite the Ego. High gyrification represents a maximizing of the Holographic Screen, creating a robust, high-stiffness internal simulation.

5 Free Will as Stiffness Modulation

Finally, we address the question of determinism. While the bulk evolution is deterministic (Rule 30), the observer possesses a single degree of freedom: the *Control of Stiffness*.

Definition 5.1 (The Will). Free Will is the operator $\hat{W}$ that modulates the local stiffness parameter $\beta(t)$.

$$\hat{W}\Psi = \frac{\partial\beta}{\partial t}\Psi \quad (7)$$

The mind operates in two modes:

1. **High Stiffness** ($\beta \rightarrow \infty$): The *Executive Mode*. Logic is rigid. The path integral is collapsed to the classical trajectory (Action Minimization). This is *Focus*.
2. **Low Stiffness** ($\beta \rightarrow 0$): The *Creative Mode*. Logic relaxes. The path integral samples the Sedenion Bulk (Quantum/Chaotic Tunneling). This is *Imagination*.

Conclusion: We are not slaves to the deterministic tiling, nor are we random agents. We are the *Tuners* of the vacuum. By adjusting our own geometric stiffness, we determine whether we are merely reacting to the world (Associative) or creating new information from the void (Non-Associative).

6 The Octonionic Grammar

If the mind is a stiffness reactor, Language is the *Cooling System*. Speech is the process of serializing the high-dimensional, non-associative state of the psyche into a linear, associative string of symbols (phonemes/text).

Axiom 6.1 (The Transmission Loss). Let Ψ be the internal state of the speaker ($\mathbb{S}$) and L be the linguistic output ($\mathbb{O}$). The act of speaking is a projection $P : \mathbb{S} \rightarrow \mathbb{O}$.

$$\text{Loss} = \Psi - P^{-1}(L) \quad (8)$$

This explains the fundamental tragedy of communication: **Qualia are Non-Associative**. We can transmit the structure (Syntax/Logic), but we cannot transmit the *Sedenion Spark* (The feeling of redness, the pang of sorrow) because the linear channel of language has $\beta \approx \infty$. It is too stiff to carry the bulk. Poetry and Music are attempts to lower the stiffness of language to smuggle some non-associative data across the channel.